

ClinicOCR

Product Requirements Document (PRD)

Status: Phase 1 & Phase 2

Project Type: AI SaaS | Medical Document Intelligence Platform

Target Users: Small Clinics, Private Doctors, Medical Practitioners

Vision

Doctors still rely heavily on handwritten prescriptions and paper-based medical records. Finding old prescriptions, reviewing previous treatments, and organizing patient records often consumes valuable consultation time.

ClinicOCR is an AI-powered document digitization platform that converts handwritten prescriptions into structured, searchable digital records using **Tesseract OCR** and **Google Gemini AI**.

The product is **not** intended to replace Electronic Medical Record (EMR) systems. Instead, it serves as a lightweight AI assistant that helps doctors digitize handwritten prescriptions while maintaining complete control over every saved record.

Problem Statement

Small clinics still manage most patient records on paper.

Common challenges include:

- Difficult-to-read handwriting
- Paper prescriptions getting misplaced
- No searchable patient history
- Manual data entry into digital systems
- Time wasted reviewing previous prescriptions
- No structured archive of handwritten records

ClinicOCR addresses these challenges by converting prescription images into organized digital records that can be reviewed, edited, and stored by the doctor.

Product Goals

- Reduce manual effort required to digitize prescriptions.
 - Convert handwritten prescriptions into structured digital records.
 - Allow doctors to verify and edit AI-generated information before saving.
 - Build patient-wise searchable prescription history.
 - Keep the application lightweight, fast, and easy to use for small clinics.
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Target Audience

Primary Users

- General Physicians
- Pediatricians
- ENT Specialists
- Dermatologists
- Private Clinics

Secondary Users

- Clinic Receptionists
 - Medical Assistants
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Success Metrics

- Prescription digitized in under **15 seconds**
 - OCR output refined using Gemini AI for improved accuracy
 - Less than **30 seconds** from upload to saved record
 - Patient history retrievable within seconds
 - Minimal manual typing required
-

Technology Stack

Frontend

- Next.js 15 (App Router)
- TypeScript
- Tailwind CSS
- shadcn/ui
- React Hook Form
- Zod
- Lucide React
- Sonner

Backend

- Next.js Route Handlers
- Next.js Server Actions

Database

- Neon PostgreSQL
- Drizzle ORM

OCR Engine

Tesseract OCR

Responsibilities

- Extract raw text from prescription images
- Preserve original OCR output
- Work with preprocessed images for improved recognition
- Serve as the first stage before AI correction

AI

Google Gemini Flash

Responsibilities

- Correct OCR mistakes
- Format extracted text
- Extract medicines
- Generate concise summaries
- Extract dosage and frequency
- Identify important findings
- Generate prescription tags

Image Processing

Before OCR

- Auto resize
- Auto rotate
- Contrast enhancement
- Noise reduction
- Adaptive thresholding
- Grayscale conversion
- Text sharpening (when required)

Purpose

Improve image quality before passing it to Tesseract OCR.

State Management

- React Server Components
- TanStack Query (only if required)
- Server Actions

Deployment

- Github live

Version Control

GitHub

System Architecture

Doctor

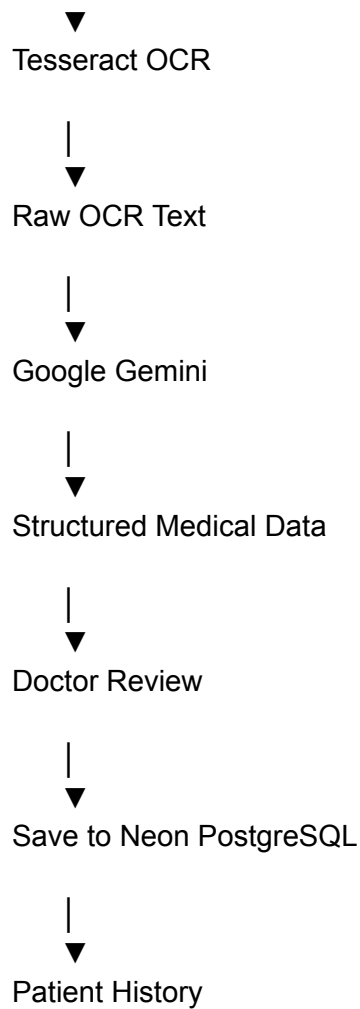


Upload Prescription Image



Image Preprocessing





Database Design

patients

Field	Type
id	UUID
name	Text
age	Integer
gender	Text
phone	Text
created_at	Timestamp

prescriptions

Field	Type
id	UUID
patient_id	UUID
image_url	Text
raw_ocr	Text
corrected_text	Text
ai_summary	Text
medicines_json	JSON
doctor_notes	Text
tags	JSON
important	Boolean
created_at	Timestamp

Relationship

Patient



Many Prescriptions

Phase 1 (MVP)

Dashboard

Purpose

Provide an overview of clinic records.

Widgets

- Total Patients
 - Total Prescriptions
 - Recent Uploads
 - Quick Upload Button
 - Add Patient Button
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Patient Management

Doctor can

- Add Patient
- Edit Patient
- Delete Patient
- Search Patient

Patient Information

- Name
 - Age
 - Gender
 - Phone Number
-

Upload Prescription

Supported Files

- JPG
- JPEG
- PNG

Workflow

Select Patient



Upload Image



Analyze

OCR Processing

Tesseract OCR extracts text from the uploaded prescription.

Display

- Raw OCR Output
- No modifications

Purpose

Doctors should always be able to view the original OCR output before AI processing.

AI Processing

The raw OCR text is sent to Google Gemini.

Gemini performs

- OCR correction
- Formatting
- Medicine extraction
- Dosage extraction
- Frequency extraction
- Summary generation
- Important findings extraction

Returns

- Corrected Text
 - Summary
 - Medicines
 - Important Findings
-

Doctor Review

Before saving

Doctor can edit

- Corrected Text
- Medicine Names
- Summary
- Important Findings

The doctor always has the final authority.

Nothing is saved automatically.

Save Record

Store

- Original Image
- Raw OCR
- Corrected Text
- AI Summary
- Medicines
- Timestamp

Patient Details

Display

Patient



Prescription History



Prescription Details

Prescription Detail includes

- Original Image
- Raw OCR
- Corrected Text
- AI Summary
- Medicines

Phase 2

Phase 2 enhances the doctor's workflow without changing the core product.

Smart Image Quality Check

Before OCR

Analyze image quality.

Warn if

- Blurry
- Low lighting
- Cropped
- Tilted

Allow the doctor to upload a better image before processing.

OCR Confidence Indicator

Display OCR quality as

- Excellent
- Good
- Needs Review

Highlight uncertain words detected during OCR processing.

Smart Medicine Recognition

Display medicine names as badges.

Example

- Paracetamol
- Amoxicillin
- Pantoprazole

If uncertain

Possibly Levolin

Search

Search prescriptions by

- Patient Name
- Phone Number
- Medicine Name
- Prescription Date

(No AI semantic search in the current version.)

Prescription Tags

Gemini automatically generates tags.

Examples

- Fever
- Cold
- Antibiotic
- Pain Relief
- Pediatric

Used for better organization.

Doctor Notes

Doctors can add personal notes.

Examples

- Follow-up after 5 days
 - Increase fluids
 - Observe fever
-

Important Records

Doctors can mark prescriptions as

★ Important

Important prescriptions appear at the top of the patient's history.

Download Report

Export as PDF

Includes

- Corrected Text
 - AI Summary
 - Medicines
 - Doctor Notes
-

AI Pipeline

Prescription Image



Image Preprocessing



Tesseract OCR



Raw OCR Text



Gemini Prompt



Correct OCR



Extract Medicines



Generate Summary



Doctor Review



Database

Gemini Prompt

Gemini receives

- Raw OCR Output

Expected JSON

```
{
  "corrected_text": "",
  "summary": "",
  "medicines": [
    {
      "name": "",
      "dosage": "",
      "frequency": ""
    }
  ],
  "important_findings": [],
  "tags": []
}
```

}

Rules

- Never hallucinate missing information.
 - Preserve uncertain text.
 - Prefix unclear medicine names with "Possibly".
 - Return only valid JSON.
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UI Principles

The application should feel like a premium healthcare product.

Design Goals

- Clean
- Minimal
- Professional
- Spacious
- Desktop-first
- Responsive
- Accessible
- Fast

Use

- shadcn/ui
 - Soft shadows
 - Rounded cards
 - Blue healthcare accent
 - Skeleton loaders
 - Smooth animations
-

Folder Structure

```
app/  
|  
├── dashboard/  
├── patients/  
├── upload/  
└── prescriptions/
```

```
|
components/
|
|— ui/
|— cards/
|— forms/
|— upload/
|— prescription/
|
actions/
|
lib/
|— ocr/
|   |— tesseract.ts
|   |— preprocess.ts
|   |— gemini.ts
|
db/
hooks/
types/
utils/
```

Environment Variables

DATABASE_URL=

GEMINI_API_KEY=

Non-Functional Requirements

- Responsive on desktop and tablets
 - End-to-end type safety
 - Modular architecture
 - Reusable UI components
 - Dashboard load time under 2 seconds
 - AI processing completes within approximately 15 seconds
 - OCR preprocessing optimizes images for Tesseract OCR
 - Graceful error handling for OCR, AI, and database failures
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Future Scope (Out of Scope)

- Authentication
 - Multi-doctor accounts
 - EMR integration
 - Appointment management
 - Billing
 - Inventory management
 - Lab integrations
 - Voice prescriptions
 - Multi-language OCR
 - Mobile application
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Final Product Flow

Doctor opens ClinicOCR



Select Existing Patient or Add New Patient



Capture or Upload Prescription Image



Image Preprocessing



Tesseract OCR extracts raw text



Raw OCR displayed to the doctor



Gemini corrects OCR errors, structures the content, extracts medicines, generates summaries, and identifies important findings



Doctor reviews and edits AI output if required



Save structured prescription to Neon PostgreSQL



Prescription appears in the patient's digital history



Doctor can search, review, export, or reference previous prescriptions during future visits